Task 6 Report
Hyperboloid Cooling Tower Optimization

Group Project 2026

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Contents

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Chapter 1

Objective

Minimize cooling-tower lateral shell area using SA, PSO, and BFGS while enforcing fixed target volume and scenario-specific constructability constraints.

The study compares stochastic and deterministic optimization behavior across eight structurally different problem setups while keeping common engineering requirements (capacity and realistic shape) explicit in the objective.

Chapter 2

Geometry Model

The tower is modeled as a stack of frustums.

- Frustum area: $A_i = \pi \cdot (r_{i-1} + r_i) \cdot \sqrt{(r_i - r_{i-1})^2 + h_i^2}$
- Frustum volume: `V_i = (pi * h_i / 3) * (r_{i-1}^2 + r_{i-1}*r_i + r_i^2)`
- Tower totals: $A = \sum(A_i)$, $V = \sum(V_i)$

Lateral shell area is optimized (top and bottom caps excluded), which directly corresponds to shell-construction material for fixed end radii.

Analytic gradients were used for both `A` and `V`, and for all penalty terms, so BFGS receives exact first-order information.

2.1 Design Variables by Scenario Type

- `radii` mode: optimize interior radii $r1 \ldots r_{m-1}$ while ring heights are fixed from `z` levels.
- `heights` mode: optimize segment heights `h1..hm` while all ring radii are fixed.
- `joint` mode: optimize both interior radii and all heights simultaneously.

Chapter 3

Optimization Methods

3.1 Simulated Annealing (SA)

SA performs single-solution stochastic search. At each evaluation, a local perturbation is proposed and accepted if it improves objective, or probabilistically accepted otherwise using the temperature schedule.

This supports escape from local minima early in search and gradual exploitation as temperature cools.

3.2 Particle Swarm Optimization (PSO)

PSO evolves a population of particles with velocity updates using inertia, cognitive pull to personal best, and social pull to global best.

It is typically strong on global exploration and multimodal landscapes, but can require many evaluations to satisfy strict constraints.

3.3 BFGS (Deterministic Quasi-Newton)

BFGS updates an inverse-Hessian approximation and computes descent directions with line search, yielding rapid local convergence when gradients are informative.

It is efficient in evaluation count but can be sensitive to initialization and non-convex penalties.

Chapter 4

Penalized Objective and Geometrical Constraints

The optimized scalar objective is $J(x) = A(x) + P_{volume} + P_{height} + P_{shape} + P_{smooth} + P_{bounds} + P_{ratio}$ (terms enabled per scenario).

4.1 Constraint Terms and Their Function

- P_{volume}: enforces target cooling capacity by penalizing relative deviation from target volume.
- P_{height}: enforces total tower height in scenarios where heights vary, preserving comparable overall structure.
- P_{shape}: enforces hyperboloid-like monotonic contraction to neck and expansion above neck.
- P_{smooth}: penalizes second differences in radii to avoid oscillatory/sawtooth shell profiles.
- P_{bounds}: hinge penalty for design-variable bounds to keep variables in practical/constructable ranges.
- P_{ratio} (S8): limits adjacent-height ratios to promote constructability and avoid abrupt segment transitions.

4.2 Feasibility Rule

- Relative volume error must be $\leq 1 \times 10^{-3}$.
- If heights vary, relative total-height error must be $\leq 1 \times 10^{-3}$.
- If radii vary, monotonic hyperboloid violation must be $\leq 1 \times 10^{-6}$.

Chapter 5

Optimization Protocol

- Runs per algorithm per scenario: 10
- Seed offset: 0
- SA: $temp_init = 120$, $cooling_rate = 0.997$, scenario-dependent $step_size$
- PSO: $w = 0.65$, $c1 = 1.6$, $c2 = 1.7$, particles = 40 or 50 by dimension
- BFGS: $tol = 1 \times 10^{-7}$, $max_iter = 1200$
- Penalties: volume, height sum (when applicable), shape monotonicity, smoothness, bounds, ratio (S8)

Stochastic budgets are 10k evaluations for lower-dimensional setups and 16k for joint/high-dimensional setups. BFGS uses smaller budgets due to faster local convergence.

Chapter 6

Scenario Definitions

ID	**Mode**	**m**	**Target Volume**	**r0**	**rm**	**Description**
S1	radii	10	70320	39.3	27.4	Required case with bounded radii and required ring heights.
S2	radii	10	70320	39.3	27.4	Required case with wide radii range (unconstrained-like).
S3	radii	10	70320	39.3	27.4	Uniform-height radii design with bounded radii.
S4	radii	12	70320	39.3	27.4	Finer discretization (m=12) with bounded radii and smoothness.
S5	heights	10	70320	39.3	27.4	Fixed radii, optimize bounded heights.
S6	heights	10	70320	39.3	27.4	Fixed radii, optimize wide-range heights (unconstrained-like).
S7	joint	10	70320	39.3	27.4	Joint bounded optimization of radii and heights.
S8	joint	10	90000	39.3	27.4	Joint constructability-aware design with larger target volume.

Chapter 7

Cross-Algorithm Summary

Algorithm	Total Runs	Mean Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
SA	80	7814.56	5.277e-04	11500.0	0.5448	71.2%
PSO	80	8264.80	1.698e-02	11500.0	0.5803	50.0%
BFGS	80	7850.64	4.018e-04	4227.9	0.2061	72.5%

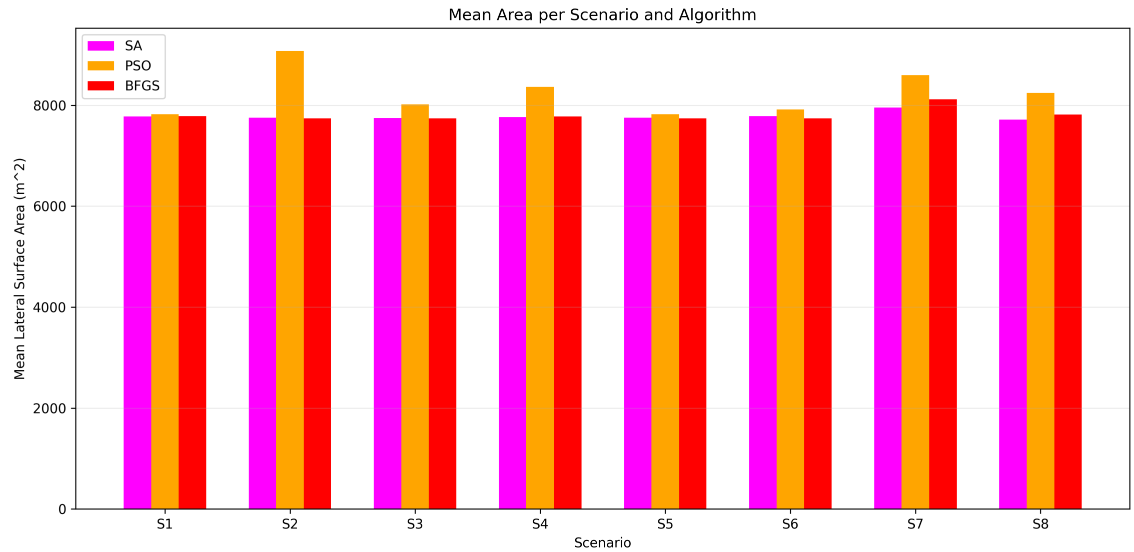

Figure 7.1: Cross-scenario area comparison

Chapter 8

Scenario Results

8.1 S1

Algorithm	Mean Area	Std Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
BFGS	7782.01	0.00	1.537e-03	3302.6	0.1651	0.0%
PSO	7950.10	201.50	1.571e-02	10000.0	0.5382	20.0%
SA	7781.85	4.69	1.246e-03	10000.0	0.5088	50.0%

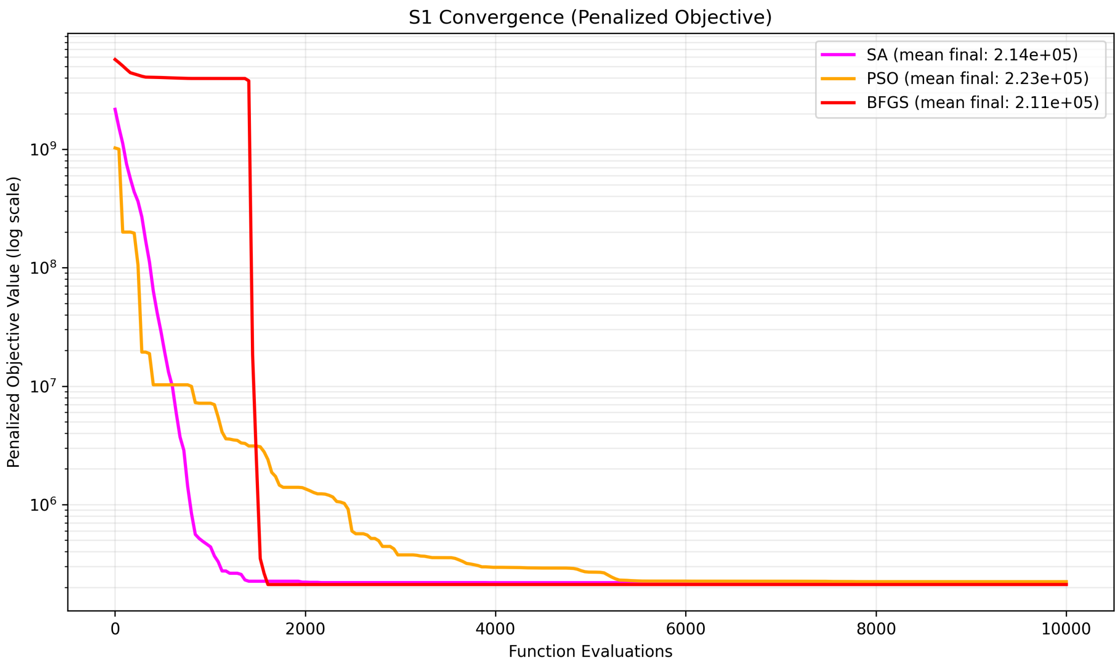

Figure 8.1: S1 convergence

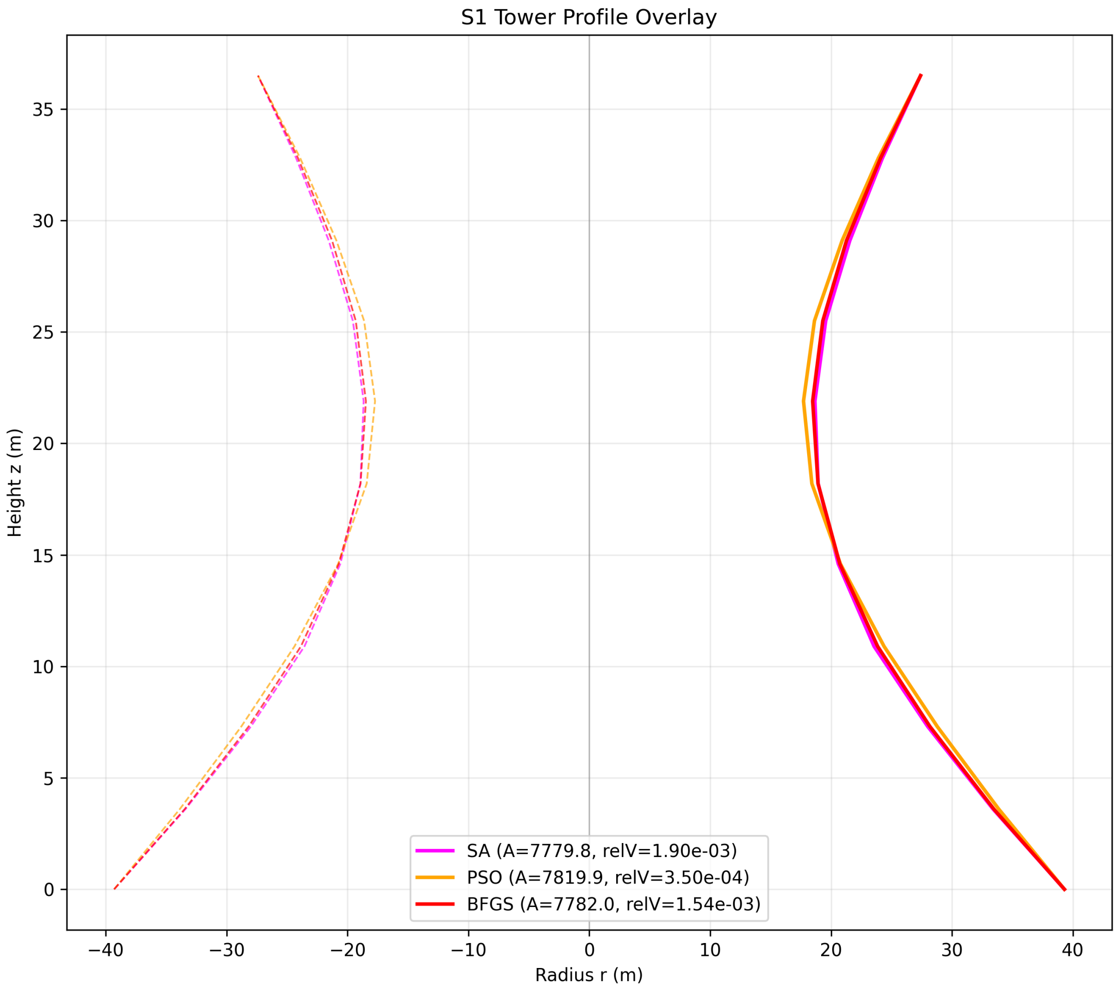

Figure 8.2: S1 profile

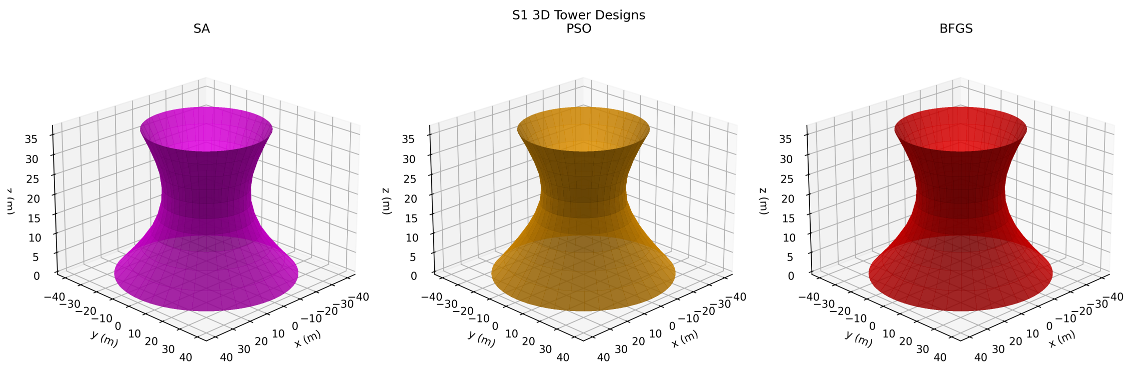

Figure 8.3: S1 3D towers

8.2 S2

Algorithm	Mean Area	Std Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
BFGS	8051.30	532.40	1.485e-04	3945.5	0.1574	100.0%
PSO	9352.84	524.78	6.536e-02	10000.0	0.4444	40.0%
SA	7753.26	9.27	5.492e-05	10000.0	0.4173	100.0%

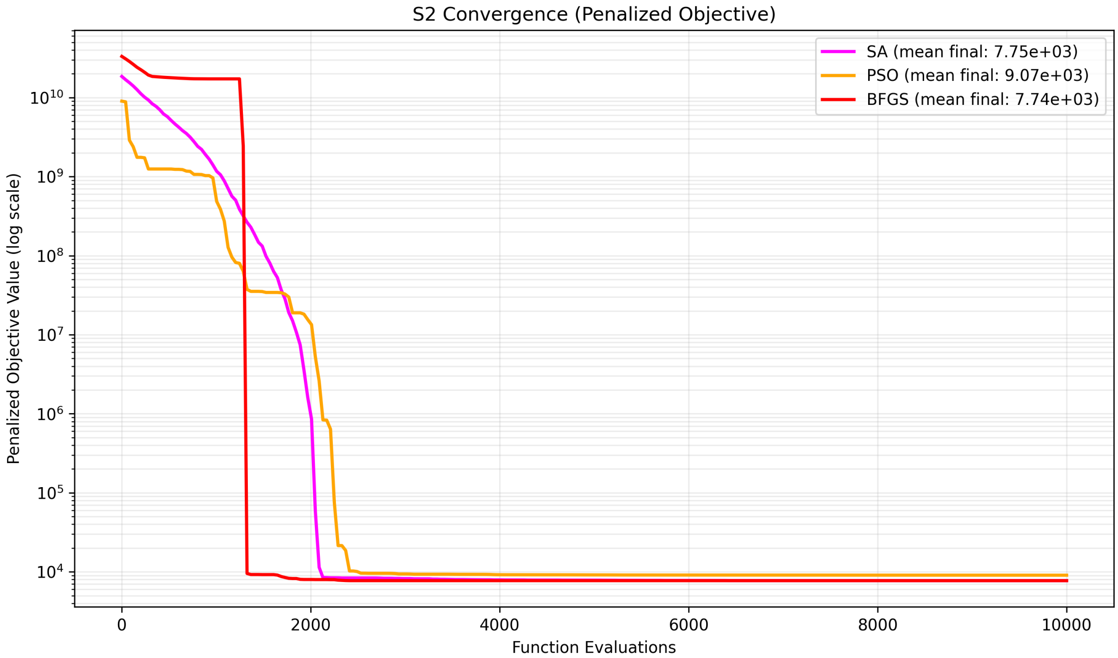

Figure 8.4: S2 convergence

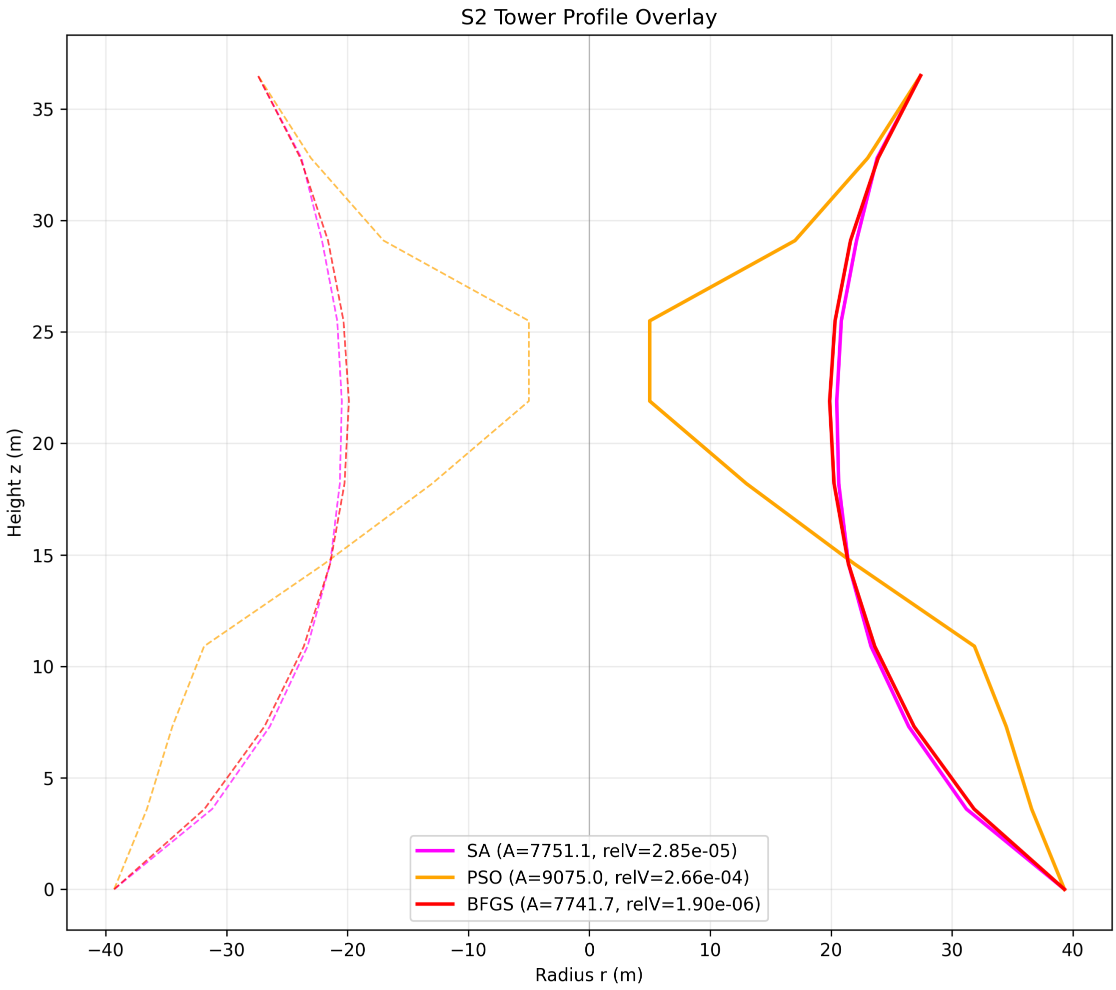

Figure 8.5: S2 profile

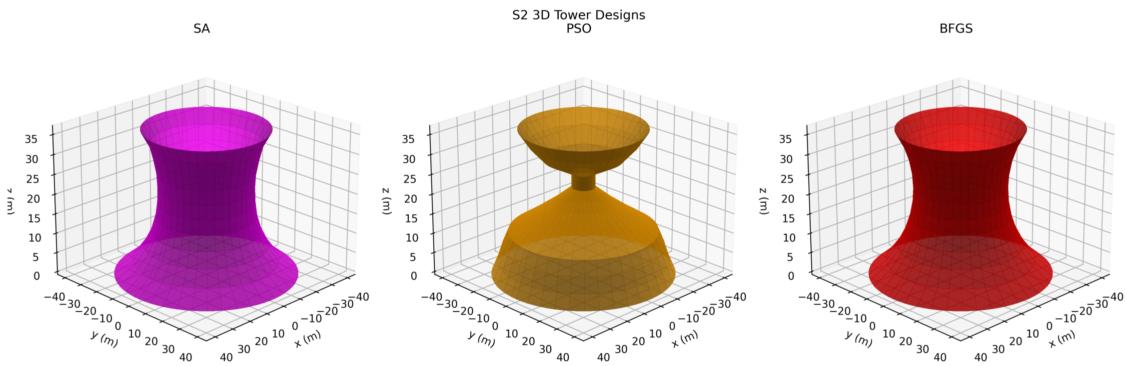

Figure 8.6: S2 3D towers

8.3 S3

Algorithm	Mean Area	Std Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
BFGS	7926.31	300.48	1.541e-04	2904.7	0.1189	100.0%
PSO	8258.12	351.80	2.509e-05	10000.0	0.4407	90.0%
SA	7753.02	21.36	3.424e-05	10000.0	0.4156	100.0%

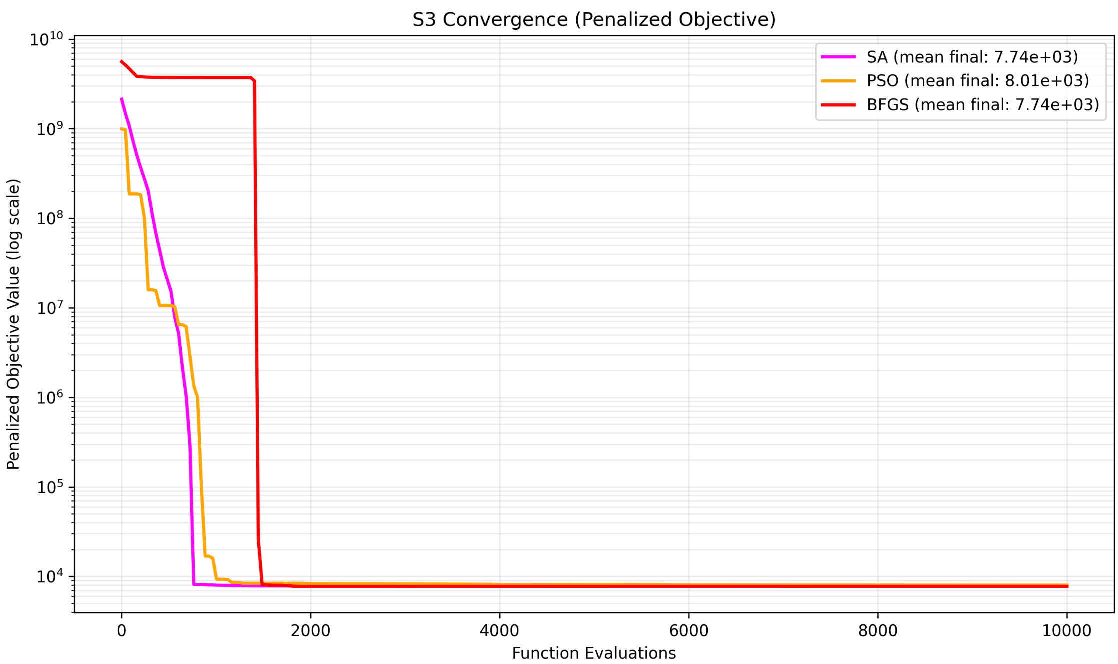

Figure 8.7: S3 convergence

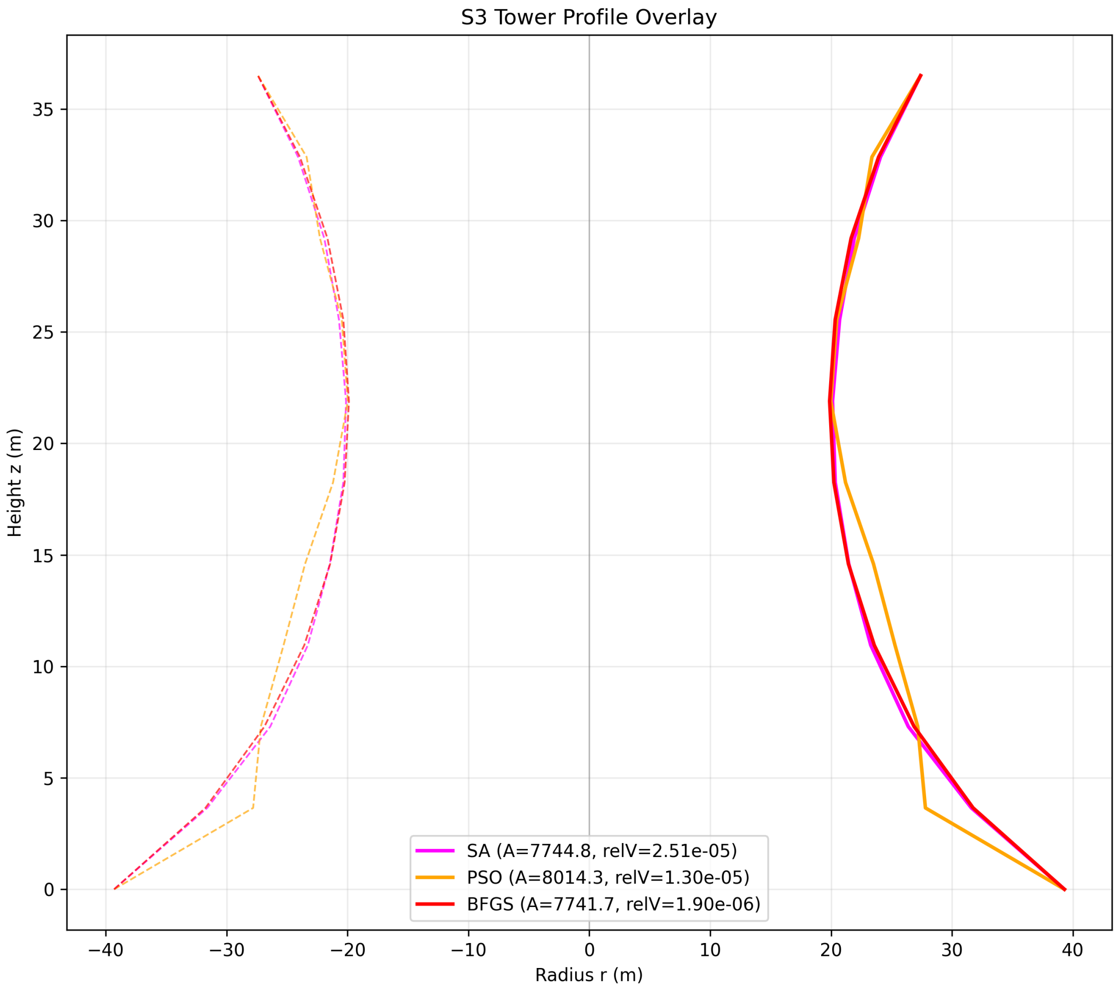

Figure 8.8: S3 profile

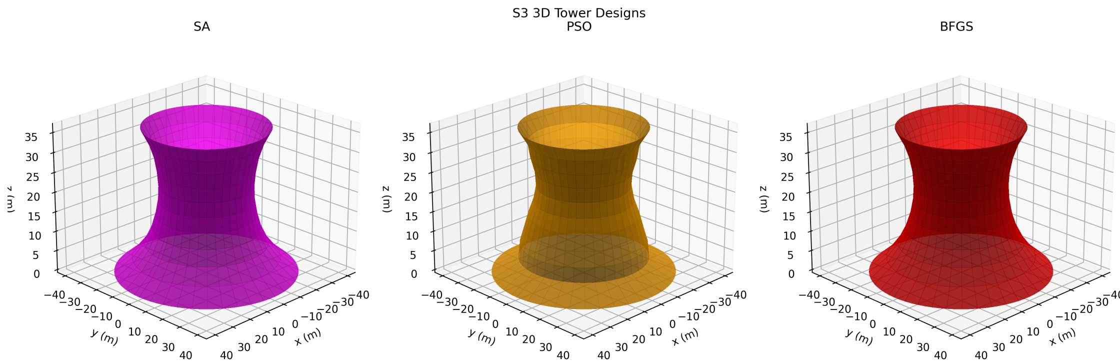

Figure 8.9: S3 3D towers

8.4 S4

Algorithm	Mean Area	Std Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
BFGS	7779.85	0.00	8.915e-04	4101.4	0.2142	100.0%
PSO	8269.55	447.60	3.272e-02	10000.0	0.5666	0.0%
SA	7783.18	14.88	1.557e-03	10000.0	0.5289	20.0%

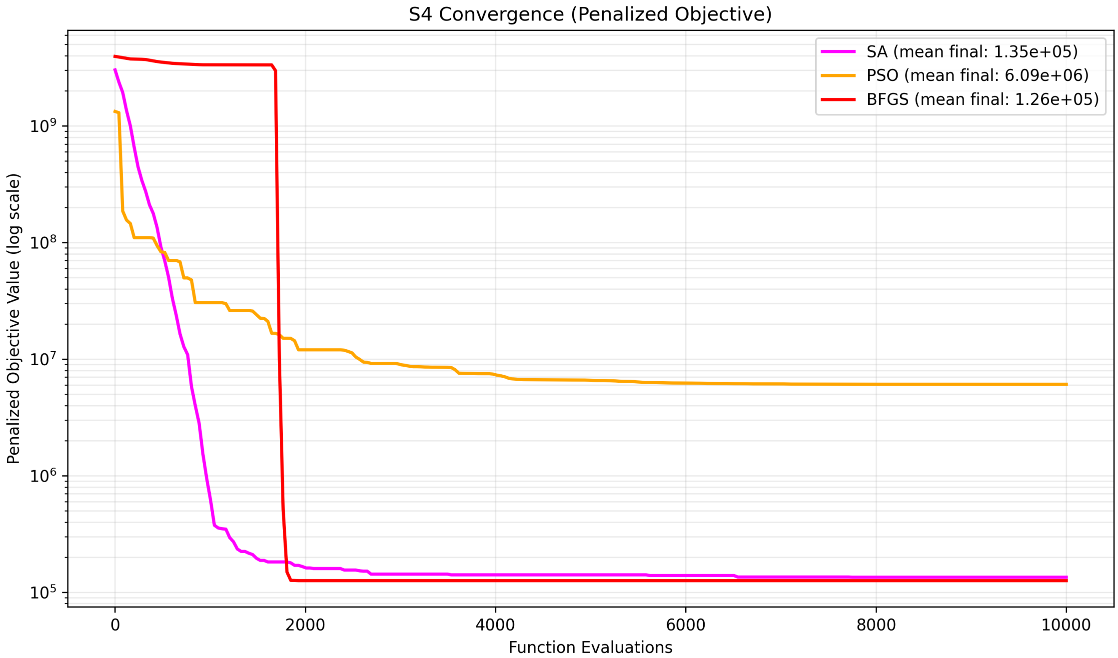

Figure 8.10: S4 convergence

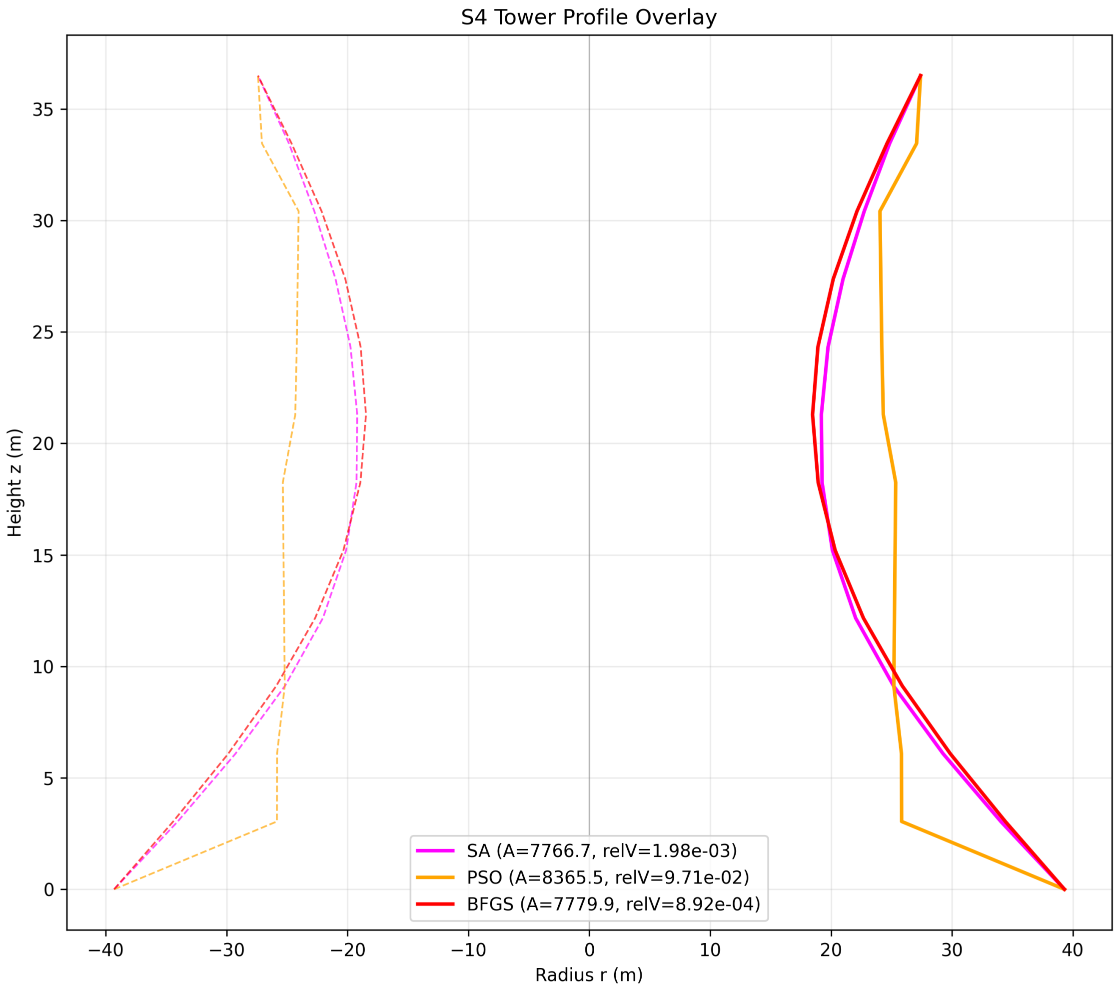

Figure 8.11: S4 profile

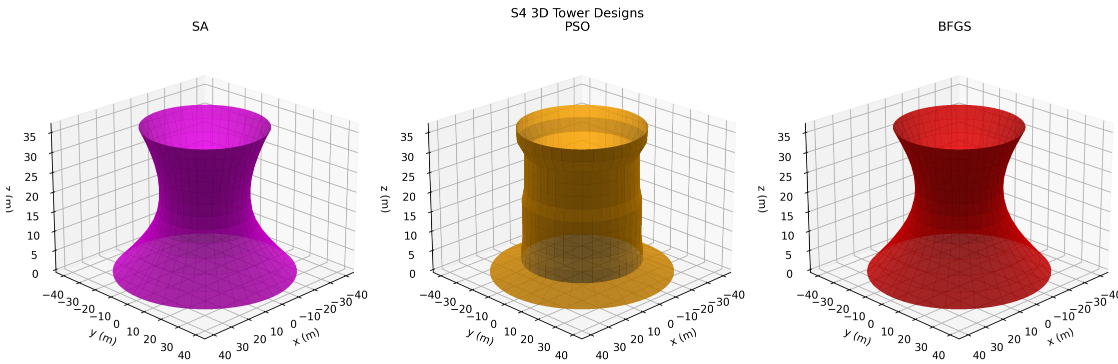

Figure 8.12: S4 3D towers

8.5 S5

Algorithm	Mean Area	Std Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
BFGS	7742.56	0.00	2.331e-06	4164.2	0.1605	100.0%
PSO	7796.77	32.30	1.128e-04	10000.0	0.4224	90.0%
SA	7757.25	6.61	3.902e-05	10000.0	0.3972	100.0%

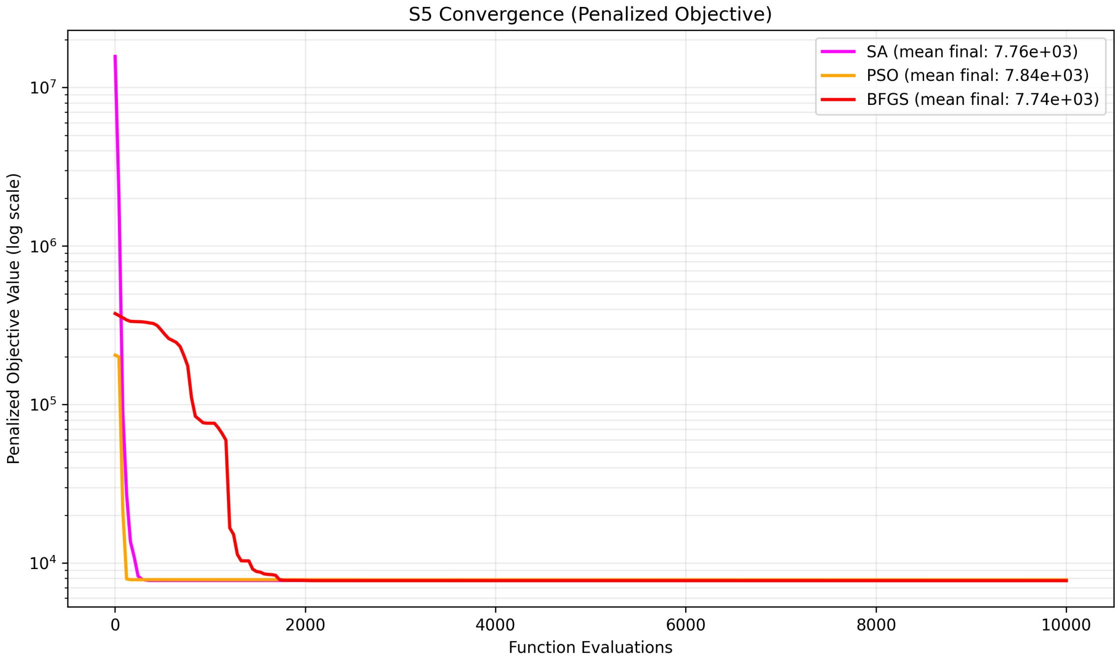

Figure 8.13: S5 convergence

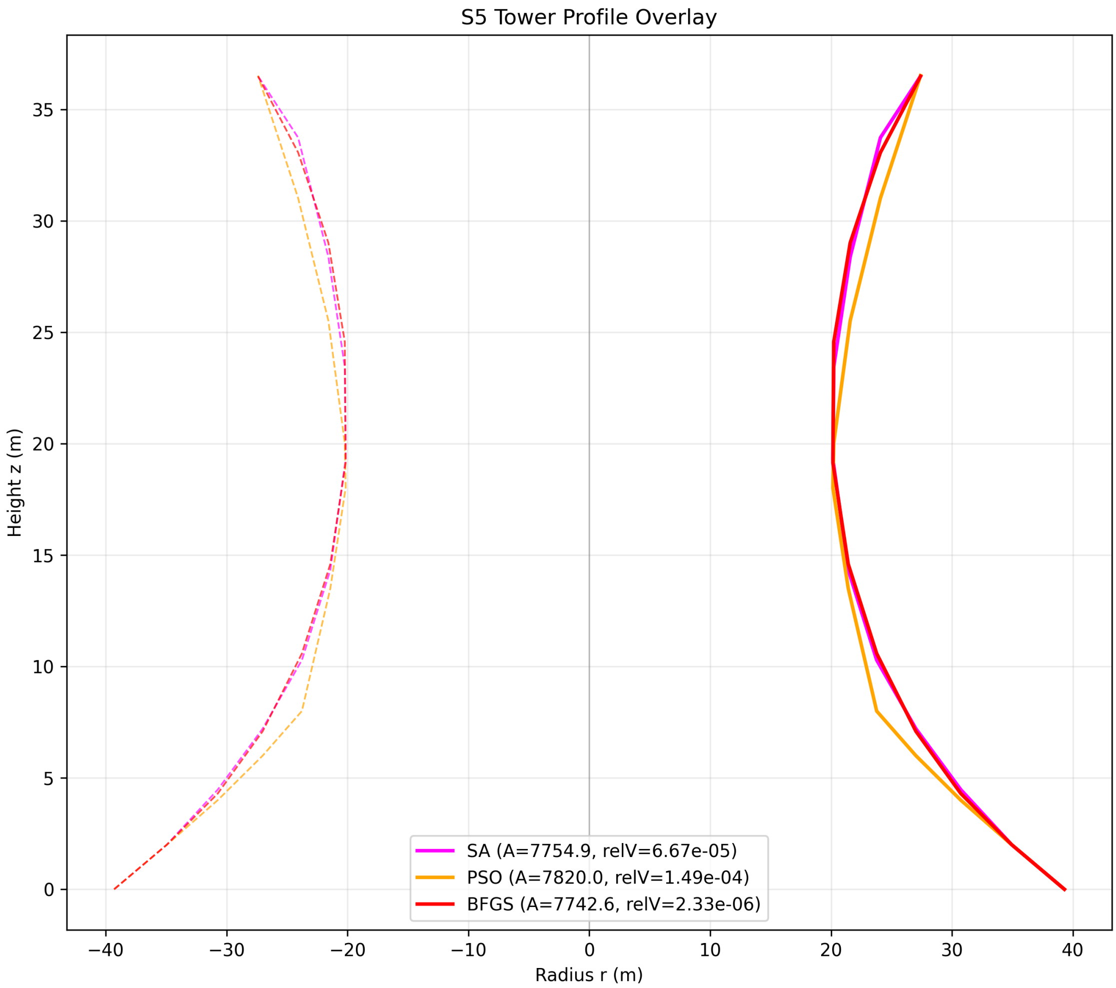

Figure 8.14: S5 profile

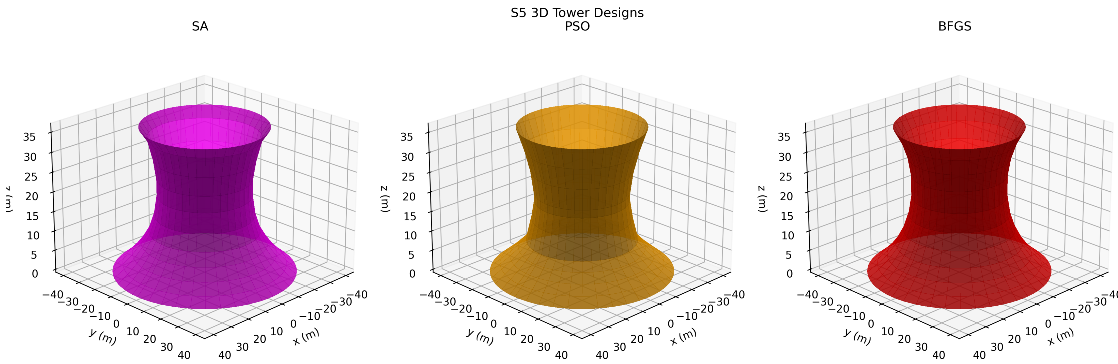

Figure 8.15: S5 3D towers

8.6 S6

Algorithm	Mean Area	Std Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
BFGS	7741.72	0.00	2.071e-06	1402.6	0.0560	100.0%
PSO	7913.68	146.68	5.923e-04	10000.0	0.4250	90.0%
SA	7825.77	33.17	6.569e-05	10000.0	0.3995	100.0%

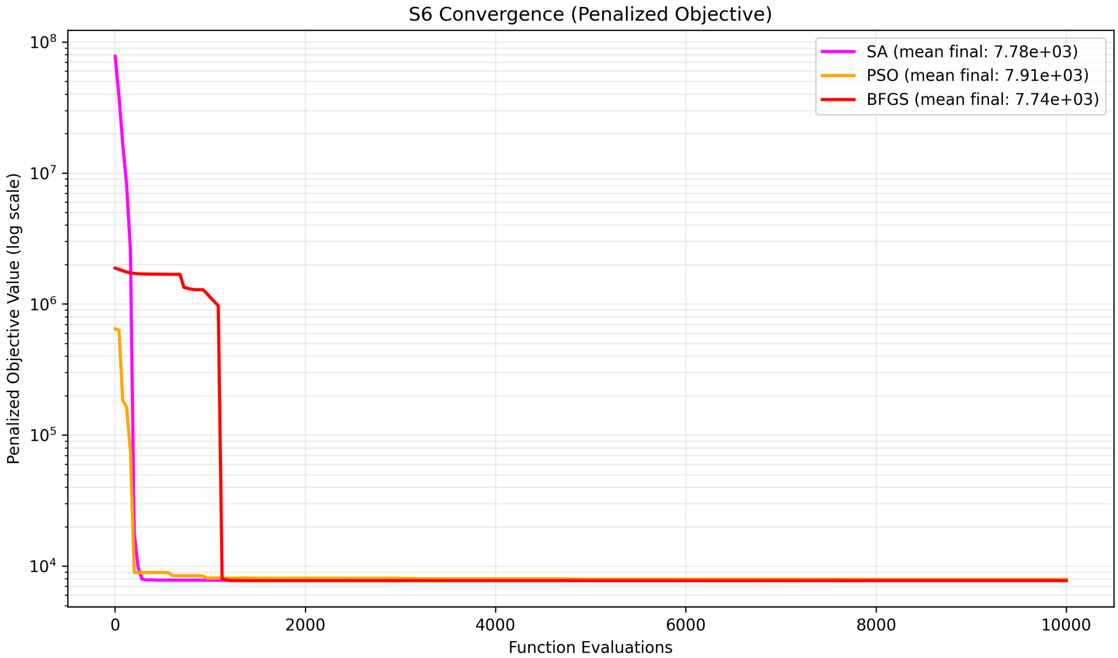

Figure 8.16: S6 convergence

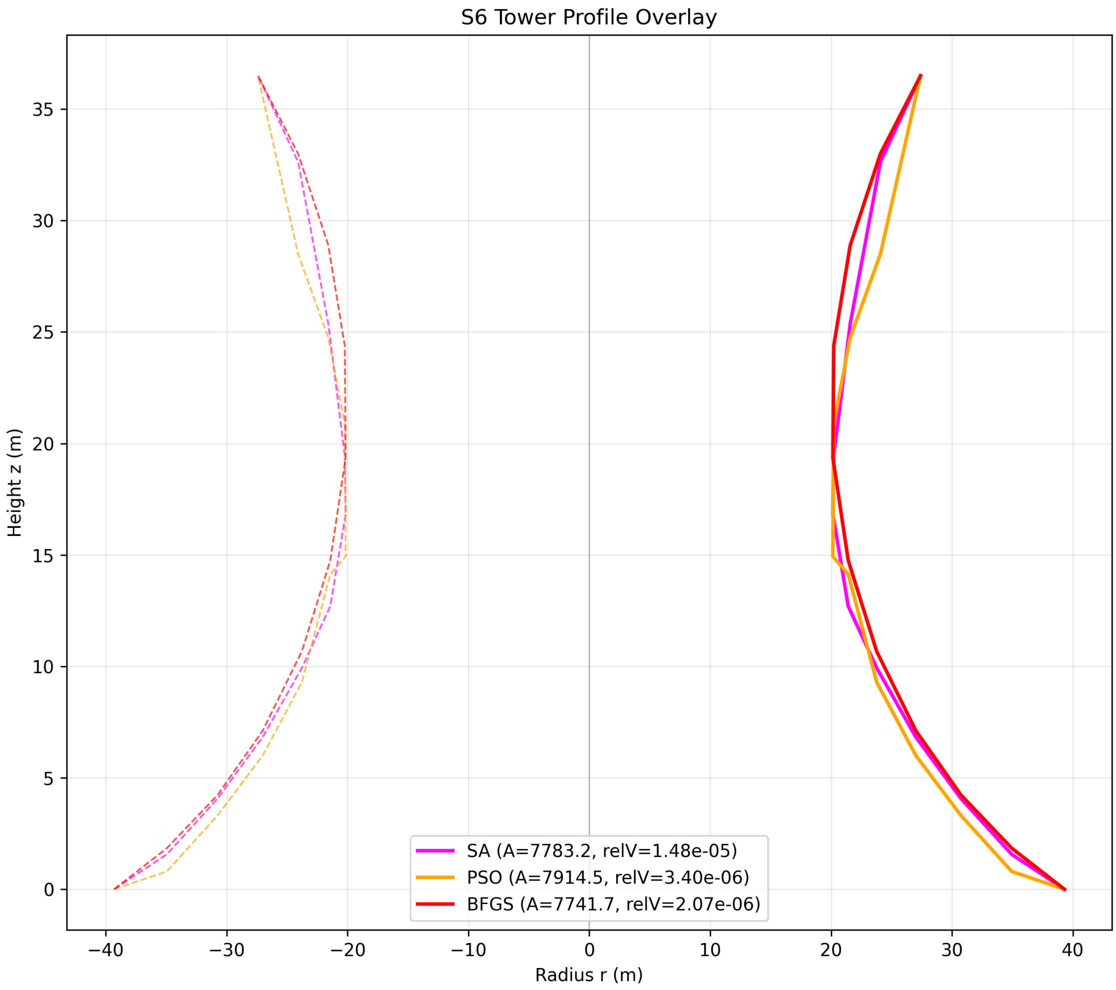

Figure 8.17: S6 profile

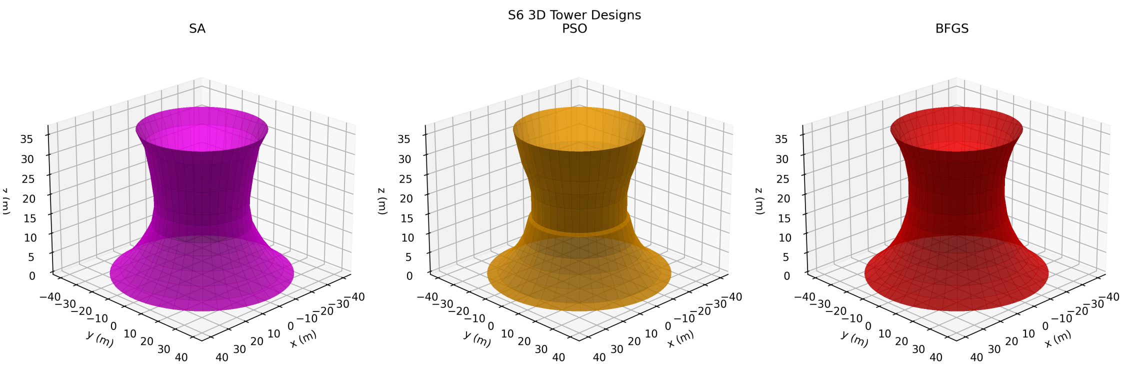

Figure 8.18: S6 3D towers

8.7 S7

Algorithm	Mean Area	Std Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
BFGS	7968.31	369.60	6.476e-05	7000.9	0.3292	80.0%
PSO	8608.62	174.56	4.144e-04	16000.0	0.7790	70.0%
SA	8160.14	281.40	1.305e-04	16000.0	0.7315	100.0%

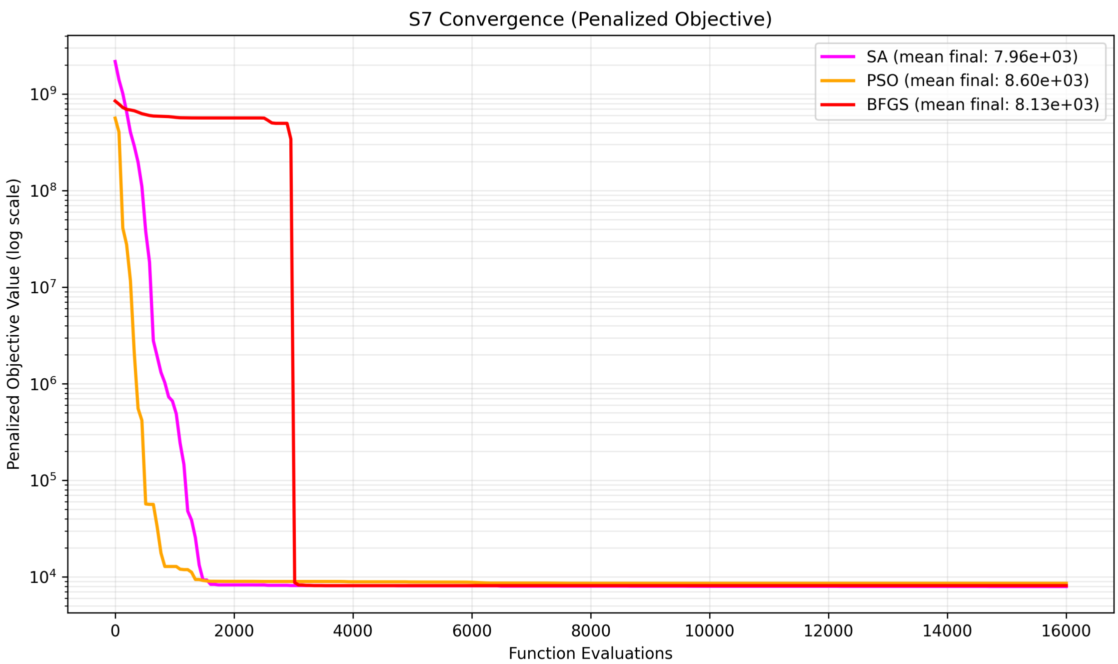

Figure 8.19: S7 convergence

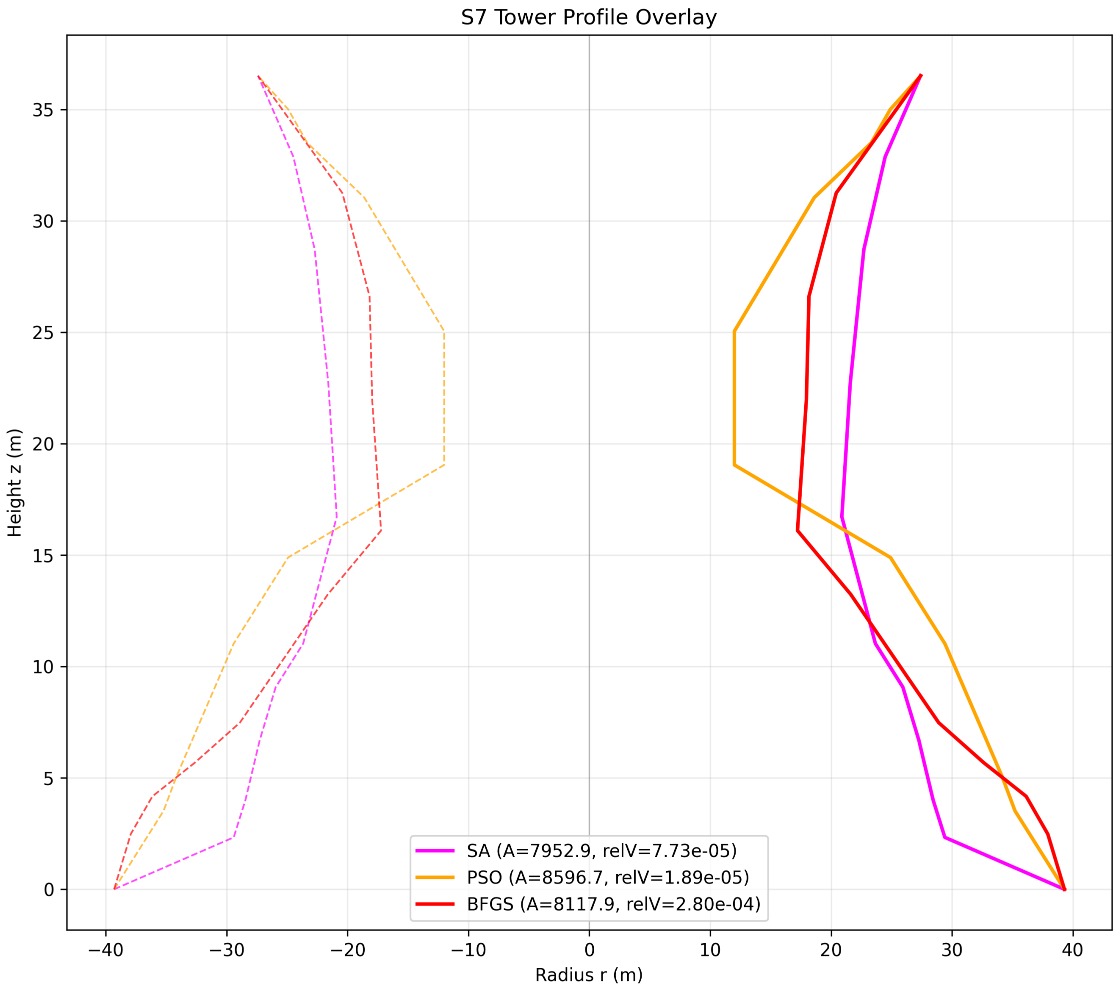

Figure 8.20: S7 profile

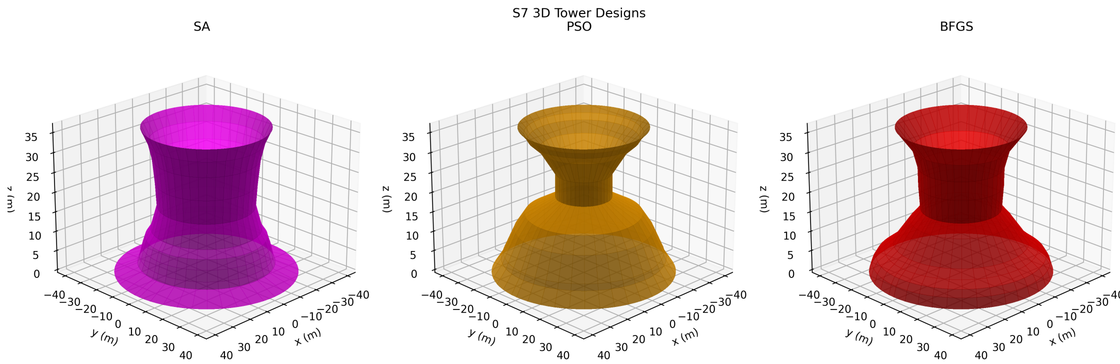

Figure 8.21: S7 3D towers

8.8 S8

Algorithm	Mean Area	Std Area	Mean Rel Vol Err	Mean Evals	Mean Time (s)	Feasibility Rate
BFGS	7813.08	0.46	4.146e-04	7001.0	0.4476	0.0%
PSO	7968.67	348.77	2.087e-02	16000.0	1.0261	0.0%
SA	7701.98	32.24	1.093e-03	16000.0	0.9600	0.0%

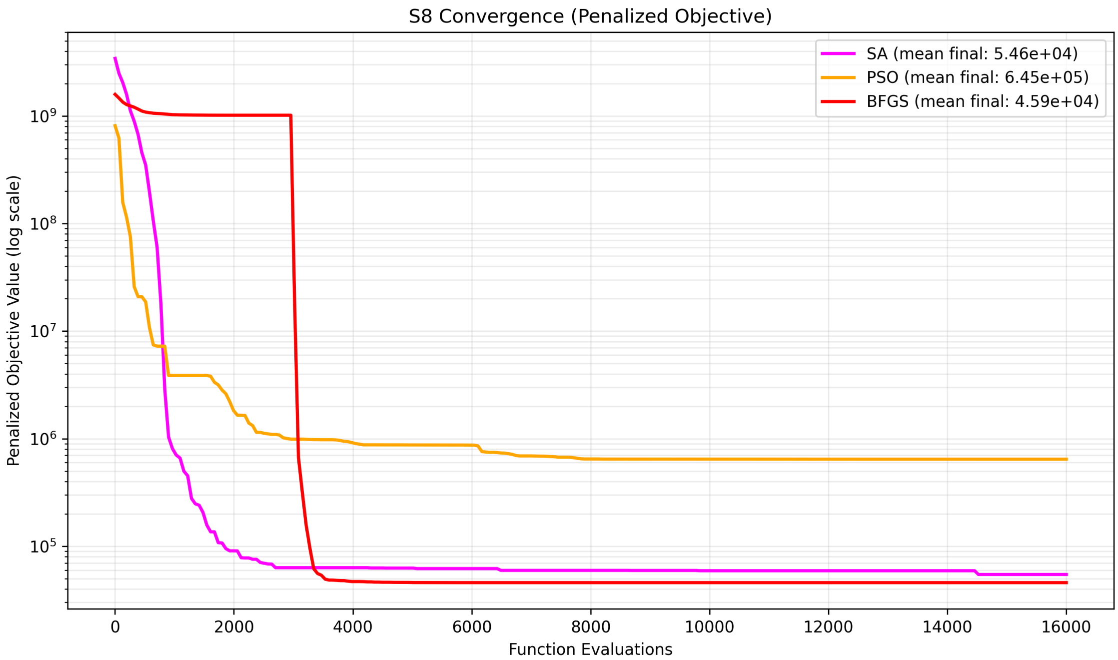

Figure 8.22: S8 convergence

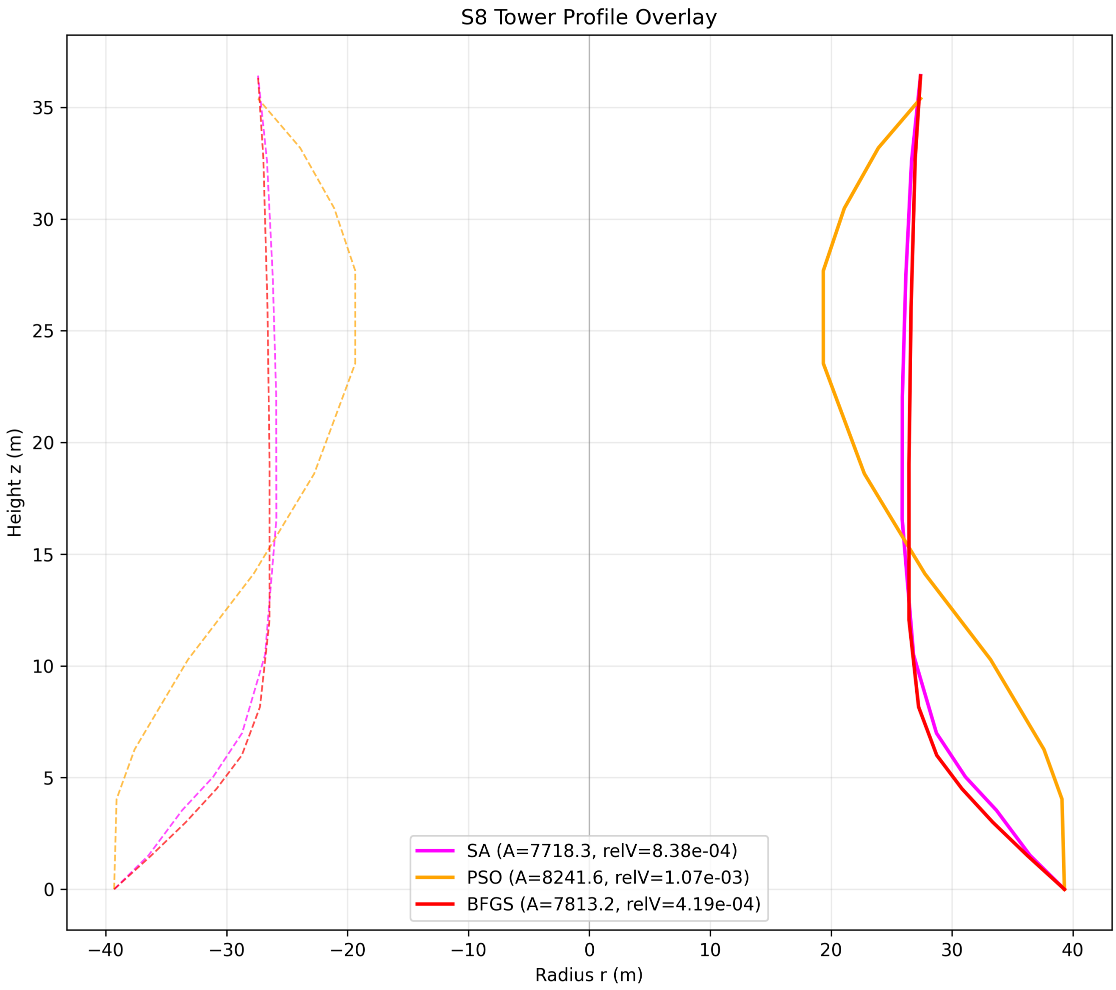

Figure 8.23: S8 profile

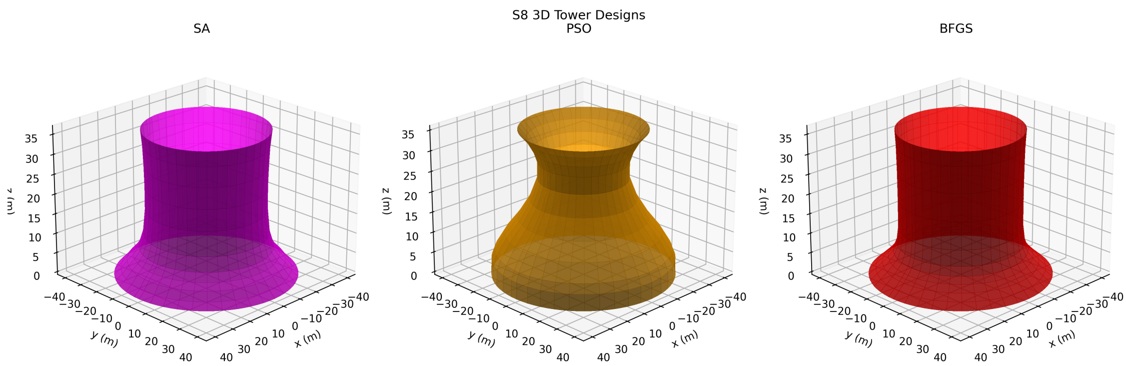

Figure 8.24: S8 3D towers

Chapter 9

Key Findings

- S1: best feasibility-area tradeoff = SA (feasibility 50.0%, mean area 7781.85 m^2).
- S1: lowest mean evaluation count = BFGS (3302.6 evals).
- S2: best feasibility-area tradeoff = SA (feasibility 100.0%, mean area 7753.26 m^2).
- S2: lowest mean evaluation count = BFGS (3945.5 evals).
- S3: best feasibility-area tradeoff = SA (feasibility 100.0%, mean area 7753.02 m^2).
- S3: lowest mean evaluation count = BFGS (2904.7 evals).
- S4: best feasibility-area tradeoff = BFGS (feasibility 100.0%, mean area 7779.85 m^2).
- S4: lowest mean evaluation count = BFGS (4101.4 evals).
- S5: best feasibility-area tradeoff = BFGS (feasibility 100.0%, mean area 7742.56 m^2).
- S5: lowest mean evaluation count = BFGS (4164.2 evals).
- S6: best feasibility-area tradeoff = BFGS (feasibility 100.0%, mean area 7741.72 m^2).
- S6: lowest mean evaluation count = BFGS (1402.6 evals).
- S7: best feasibility-area tradeoff = SA (feasibility 100.0%, mean area 8160.14 m^2).
- S7: lowest mean evaluation count = BFGS (7000.9 evals).
- S8: best feasibility-area tradeoff = SA (feasibility 0.0%, mean area 7701.98 m^2).
- S8: lowest mean evaluation count = BFGS (7001.0 evals).
- Feasibility bottleneck scenarios: S8 (all tested methods yielded 0% feasible runs under current constraints).

Chapter 10

Discussion and Conclusions

Across all scenarios, \textbf{BFGS} achieved the highest overall feasibility (72.5%).

\textbf{BFGS} required the fewest evaluations on average (4227.9), confirming its efficiency for local convergence.

By raw mean area across all runs, \textbf{SA} produced the lowest average shell area (7814.56 m^2), but this must be interpreted jointly with feasibility rates.

Result patterns are consistent with expected method behavior: BFGS is computationally efficient and strong when gradients and local curvature align with feasible basins; SA is often robust under hard nonlinear penalties because probabilistic acceptance helps transition between constrained basins; PSO explores broadly but can underperform feasibility when penalty landscapes are steep and high-dimensional.

For practical engineering usage in this problem family, the recommended workflow is a hybrid strategy: use SA/PSO for global search and feasibility discovery, then warm-start BFGS for rapid refinement of the best feasible candidate.

Scenario S8 remains the principal bottleneck under the current penalty/budget setup; this indicates that constructability and enlarged-volume requirements are jointly tight. Future refinement should focus on adaptive penalty scheduling, better initialization heuristics for joint variables, and potentially increased stochastic budget for S8.

Chapter 11

Deliverables

- `results/raw_runs.csv`
- `results/scenario_summary.csv`
- `results/algorithm_summary.csv`
- `results/scenario_definitions.json`
- `results/figures/*.png`